

Gifts

You are hosting a party with n guests but have prepared only m gifts, where $m < n$. To decide who should receive a gift, you assign each guest a tier number t_i ($1 \leq t_i \leq 10^9$). The lower the tier number, the closer the guest is to you. Guests with a tier number of 1 are your closest friends.

You will distribute the gifts based on the tier numbers:

- First, give gifts to all guests in tier 1.
- Then, proceed to tier 2, tier 3, and so on.
- If there are not enough gifts for all guests in a tier, prioritize guests based on their arrival order. For simplicity, guests arrive in the order of their indices; that is, guest i arrives before guest j if $i < j$.

Determine which guests will receive a gift.

Input Format

- The first line contains two integers n and m ($1 \leq m < n \leq 10^5$) — the number of guests and the number of gifts.
- The second line contains n integers $t_1, t_2, \dots, t_n$ ($1 \leq t_i \leq 10^9$) — the tier numbers of the guests.

Output Format

Output a single line containing n integers $x_1, x_2, \dots, x_n$, where $x_i = 1$ if guest i receives a gift, and $x_i = 0$ otherwise.

Sample Input

```
8 6
3 1 4 1 5 9 2 5
```

Sample Output

```
1 1 1 1 1 0 1 0
```

Explanation

- **Guests 2 and 4** are in tier 1 and both receive gifts.
- **Guest 7** is in tier 2 and receives a gift.
- **Guest 1** is in tier 3 and receives a gift.
- **Guest 3** is in tier 4 and receives a gift.
- **Guest 5 and 8** is in tier 5 but there is only one gift left. Since $5 < 9$, **Guest 5** receives a gift but not **Guest 8**.
- **Guest 6** is in tier 9 and does not receive a gift.

Constraints

- $1 \leq m < n \leq 10^5$,
- $1 \leq t_i \leq 10^9$ for all $1 \leq i \leq n$